The Slow Architecture of Coral

Teacher copy (everything in the student copy, plus answers and teaching notes)

The article, marked for rhetorical devices (one color throughout):

1 A coral reef is built out of a mineral the animal manufactures one tiny particle at a time. The mineral is aragonite, a form of calcium carbonate, the same compound as chalk but a different crystal arrangement than the calcite that makes up most limestone. A polyp does not pull finished crystal from the sea. It assembles soft, formless beads of calcium carbonate first, each about 400 nanometers across, and holds them in its tissue for hours before they harden into aragonite. That detour through a formless phase is roughly a hundred times faster than growing the crystal ion by ion straight out of seawater.

2 To make the mineral at all, the polyp has to change the chemistry of a sliver of space. Between its own cells and the skeleton it has already laid down, it pumps hydrogen ions out of that gap. Removing the hydrogen tips the water's balance toward carbonate ions, and once enough carbonate has gathered there, calcium and carbonate can lock together into solid calcium carbonate. **[Logos]** A separate set of proteins, secreted by the coral and called the soluble organic matrix, folds itself into ordered layers that steer which crystal forms, hold the formless beads stable, and suppress rival minerals the chemistry would otherwise allow. The skeleton grows the way it does because the animal keeps editing the conditions under which it grows.

3 None of this would sustain a reef without a tenant. Inside the polyp's tissue live single-celled algae, photosynthetic dinoflagellates called zooxanthellae, packed at more than a million cells in every square centimeter. **[Imagery]** The coral breathes out carbon dioxide and water; the algae take both and run photosynthesis; and the algae hand as much as ninety percent of what they make back to the coral that houses them. The lipids alone tell the story of the bargain: of the wax esters and fats the algae produce, about seventy-five percent goes to the host and only eight percent stays behind. It is an arrangement that lets coral flourish in tropical water so nutrient-poor it should be close to a desert.

4 Light feeds the skeleton as well as the animal. **[Telegraphic sentence]** When the algae photosynthesize they pull carbon dioxide out of the surrounding tissue, the local water turns less acidic, carbonate becomes easier to come by, and the coral lays down skeleton faster in the light than in the dark. The effect has a name, light-enhanced calcification, and for a long time the obvious reading was that the algae simply fed the process. The picture turns out to be less tidy. Under blue light, calcification can hold near its normal pace even when the algae's photosynthesis is turned down, which points to blue-light receptors in the coral itself triggering the building through some signal we do not yet fully trace.

5 The building runs on a clock, or several clocks at once. Through the day the algae convert sunlight into chemical energy; through the night the colony shifts toward feeding on the water and toward repair. **[Parallelism]** Many stony corals open and close their polyps on a daily beat set by light, and time their spawning to particular phases of the moon. The skeleton itself keeps the time. Its fibers are laid down in rhythmic pulses, repeated deposits of acidic proteins building a layered structure pulse on pulse, the way a

tree lays a ring. [Simile] Genus by genus the pace differs, somewhere between a third of a centimeter and two centimeters a year, depending on warmth and food and how clear the water is.

6 Because that layering is steady and the chemistry records its surroundings, a single old colony becomes an archive. Mounding corals of the genus Porites can live for centuries; their annual density bands store, in order, the seawater temperature and salinity and acidity they grew through, month by month and year by year. [Personification] Cores drilled from reefs reach back across the ten thousand years of the Holocene. There is a limit worth keeping in view. The same warmth that the bands faithfully record can also break the partnership the colony is built on: pushed past the hottest water it normally meets, a coral will digest and cast out the very algae that feed it, and the long patient record goes on being written by the thing that can end it.

Devices, and why they are here:

Logos (paragraph 2): “Removing the hydrogen tips the water's balance toward carbonate ions, and once enough carbonate has gathered there, calcium and carbonate can lock together into solid calcium carbonate.”
The sentence walks the reader through the actual causal chain of calcification (remove hydrogen, carbonate rises, calcium and carbonate bond) so belief comes from following the mechanism, not from being told it is impressive. In the Long Look, accurate cause-and-effect is the ethos move.

Imagery (paragraph 3): “Inside the polyp's tissue live single-celled algae, photosynthetic dinoflagellates called zooxanthellae, packed at more than a million cells in every square centimeter.”
Concrete, measurable sensory detail (single-celled algae living inside tissue, more than a million cells per square centimeter) makes the scale of the symbiosis something the reader can picture rather than an abstraction, which is how this register earns awe through precision instead of adjectives.

Telegraphic sentence (paragraph 4): “Light feeds the skeleton as well as the animal.”
A short flat declarative snaps against the long accumulating sentence that follows it, the signature Long Look rhythm. Its brevity states the paragraph's finding plainly and lets the slower technical sentence carry the qualification.

Parallelism (paragraph 5): “Through the day the algae convert sunlight into chemical energy; through the night the colony shifts toward feeding on the water and toward repair.”
The matched 'Through the day... through the night...' clauses set the colony's two shifts side by side in balanced structure, enacting the daily cycle in the sentence's own shape. The repeated phrase sits at the head of each clause but the construction is built on grammatical balance of the two halves.

Simile (paragraph 5): “Its fibers are laid down in rhythmic pulses, repeated deposits of acidic proteins building a layered structure pulse on pulse, the way a tree lays a ring.”
The explicit comparison using 'the way' likens the skeleton's pulsed fiber deposits to a tree laying a ring, mapping an unfamiliar biomineral process onto a familiar growth pattern so the reader grasps both the layering and the record-keeping at once.

Personification (paragraph 6): “Mounding corals of the genus Porites can live for centuries; their annual density bands store, in order, the seawater temperature and salinity and acidity they grew through, month by month and year by year.”
Treating the coal colony as an 'archive' that 'stores' and keeps a 'record' assigns a human, deliberate act of record-keeping to the skeleton, which lets the passage hand off into the closing turn where the same record-keeping carries the threat to it.

AP-style multiple choice:

1. In the fourth paragraph, the writer follows the explanation of light-enhanced calcification with the observation that 'The picture turns out to be less tidy' primarily to

- A) concede that the role of light in building the skeleton remains entirely unknown
- B) argue that earlier researchers misunderstood how coral and algae cooperate
- C) qualify the straightforward account by introducing a finding the simple version does not explain
- D) emphasize that blue light is the most important factor in coral growth

2. In context, the writer's choice to call the old colony 'an archive' whose bands 'store' temperature and salinity 'in order' best conveys which of the following about the writer's stance toward the coral skeleton?

- A) that the writer regards the reef chiefly as a resource for human study
- B) that the skeleton performs an orderly, deliberate-seeming act of keeping a record over time
- C) that the coral consciously intends to document its environment for the future
- D) that the skeleton is fragile and its record easily erased

Multiple choice answer key:

1. Answer C. The phrase pivots from the clean explanation (the algae feed the process) to evidence that complicates it: under blue light, calcification continues even when photosynthesis is reduced, pointing to receptors in the coral itself. The option claiming the role is 'entirely unknown' overstates the text, which says the mechanism is not fully traced, not that nothing is known (over-broad or extreme). The option about researchers misunderstanding imports a charge of error the passage never levels and shifts to an argumentative stance this observational piece does not take (out-of-passage import). The option naming blue light the most important factor mistakes one qualifying finding for an overall ranking the text does not make (true but not the function asked).

2. Answer B. The archive language assigns the skeleton an orderly, record-keeping function, treating it as something that methodically preserves its conditions, which is the personification the passage builds on. The 'fragile and easily erased' option contradicts the text, which stresses centuries-long persistence before the closing threat (inverted relationship). The 'resource for human study' option seizes on the mention of cores but makes the writer's interest human-centered when the stance stays on the coral's own process (right effect, wrong location). The 'consciously intends' option reads the figurative archive too literally, granting the coral deliberate human intention rather than a deliberate-seeming pattern (too literal).

Discussion questions:

1. The piece spends its first two paragraphs on chemistry the reader cannot see (formless beads, pumped hydrogen ions, steering proteins) before any mention of the algae most people associate with coral. What does the writer gain by building the skeleton's chemistry first and introducing the famous partnership only afterward?

2. The closing sentence notes that the same warmth the skeleton faithfully records is the warmth that can make a coral expel the algae that feed it. How does ending on that doubling, rather than on a summary of how coral builds, change what the reader is left holding?

Sample strong discussion responses:

1. By opening on the invisible chemistry, the writer establishes that even the part of coral nobody pictures is exact and ordered, so that when the well-known algae partnership finally arrives it reads as one more precisely engineered system rather than as the whole story. The delay also lets the reader feel the reef being assembled from the smallest scale upward, particle to mineral to colony, which matches the piece's patient accumulation and earns the wonder through detail instead of asserting it.

2. Ending on the doubling leaves the reader holding a tension the writer refuses to resolve: the colony's great strength, its faithful centuries-long record, is produced by the same process that can destroy it. Rather than closing on a tidy lesson about how coral builds, the piece turns inward on a limit, which keeps the tone observational and honest about what the record cannot protect itself from, and lets the reader sit with the irony rather than be told what to conclude.

Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):

In the passage above, the writer describes how coral constructs its skeleton and sustains its partnership with the algae that live inside it. Read the passage carefully. Then write an essay in which you analyze the rhetorical choices the writer makes to convey the precision and scale of coral as a biological system. Support your analysis with specific evidence from the text, attending to choices such as the ordering of information, the movement between short declarative sentences and long accumulating ones, figurative language, and the writer's candor about what is not yet fully understood.

Common misconceptions to watch for:

- Students may read the passage as arguing a position (for example, that climate change is destroying reefs) because the final sentence raises warmth and the loss of algae. The piece is observational, not argumentative: the closing turn names a limit and a tension rather than assigning blame or calling for action, which is why the writing prompt is rhetorical analysis (Q2) and not argument (Q3).
- Students often call the matched 'Through the day... through the night...' construction in paragraph 5 anaphora because a phrase repeats at the head of each clause. The marked device is parallelism: the move depends on the balanced two-part grammatical structure setting day against night, not merely on a word recurring at the start. Anaphora would be the same opening word driving a longer run of clauses for cumulative force.
- Students may treat the skeleton 'archive' that 'stores' a record (paragraph 6, marked as personification) as literally true, concluding the coral deliberately documents its environment. The language is figurative: it grants the skeleton a human, intentional act of record-keeping to dramatize how faithfully the bands preserve conditions, which is also what makes the closing reversal land.
- Students sometimes assume photosynthesis fully explains light-enhanced calcification. The passage is careful here: under blue light, calcification can continue even when photosynthesis is reduced, which points to light receptors in the coral itself and a signal the writer says is not yet fully traced. The candor about that uncertainty is itself a rhetorical choice, not a gap to skim past.

AP exam alignment:

This opener teaches students to analyze how an observational, expository writer builds credibility and wonder through precision rather than assertion: the ordering of information (chemistry before the famous symbiosis), the Long Look rhythm of short declaratives snapped against long accumulating sentences, and figurative moves a reader can name and defend (a simile, a personification). It pairs a Reasoning and Organization MCQ on how a qualifying observation functions with a Style MCQ on figurative diction and stance, and the Q2 prompt asks students to analyze rhetorical choices in a passage that argues nothing, distinguishing rhetorical analysis from argument.